



Interference among the neighboring sub-micron size harmonic particles

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ABSTRACT

In recent times, harmonic nanoparticles (HNPs) have gained considerable interest in many fields like nanophotonics, cell imaging, nanomedicine, etc. Unlike fluorescent particles and quantum dots, HNPs are photo-stable, non-bleaching, bio-compatible, and wavelength-flexible. Here, we explore second harmonic generation (SHG) from sub-micron size BaTiO₃ (BTO) particles. BTO is a well-known perovskite material extensively used for its nonlinear optical properties. Our investigation of out-of-focus diffraction patterns of second harmonic light has helped us know about particle assembly's nature. We have observed interference fringes at the overlapped regions of the out-of-plane SHG patterns that emerged from two, three, and even multiple adjacent particles. Our study may help in the 3D tracking of nonlinear particles in a living cell and deep tissue imaging.

1. Introduction

In the last two decades, Harmonic NanoParticles (HNPs) have emerged as highly efficient probes in many applications, from cell imaging to drug delivery, nanomedicine, cancer diagnosis, imaging in turbid media, and many more [1,2]. HNPs are typically non-fluorescent particles and emit second, third, and higher harmonics depending on their material properties. Among all the harmonics, the lowest order (second) is widely explored in many fields. Typically, three kinds of HNPs are identified, namely plasmonic [3–5], dielectric [6–8] and hybrid [9–11]. The plasmonic nanoparticles (NPs) are mostly centrosymmetric in their crystal structure, wherein nonlinear interactions happen at the surface where the symmetry breaks. As a result, harmonic emissions in plasmonic NPs strongly rely on their shape. However, plasmonic NPs suffer huge losses at optical frequencies and low damage thresholds and are rather challenging to fabricate. On the contrary, dielectric NPs rely on their bulk properties for second-harmonic generation. Dielectric NPs possess a non-centrosymmetric structure and hence the $\chi^{(2)}$ -nonlinearity, and can tolerate high laser intensity. When the dimension of the bulk crystal (non-centrosymmetric) is significantly larger than the coherence length, SHG is governed by the phase-matching condition. However, this condition becomes more relaxed for nanoparticles, where the size is much smaller than the coherence length [2]. All of these make dielectric NPs an attractive candidate for further investigation.

In contrast with the fluorescence markers and quantum dots, HNPs are photo-stable, bio-compatible, non-bleaching, and offer high contrast

against the background. Furthermore, they offer wavelength flexibility, in that they can generate second or third harmonics at any fundamental wavelength within the material's transparency range. This has opened up a new excitation range at NIR wavelengths (NIR-II: 1100–1350 nm, NIR III: 1600–1870 nm). At these spectral ranges, water absorption and scattering efficiency are comparatively low. These have favored HNPs in deep tissue imaging, and multi-photon imaging in the brain [12]. Some well-known dielectric based HNPs are ZnO, Fe(IO₃)₃, BaTiO₃, LiNbO₃, KTiOPO₄, etc [13]. In particular, BaTiO₃ (BTO) NPs have quite high optical nonlinearity. Being a perovskite material, they have also been explored extensively in solar cell technology [14]. Recently, researchers have observed enhancement in second-harmonic generation in a single BTO nanoparticle of size 200 nm with the help of Mie resonance [15]. Even, a single BTO nanoparticle down to 22 nm has been reported to emit SHG [16]. Polarization-resolved SHG imaging has enabled to reveal of the crystalline nature at the nanoscale [17]. Moreover, BTO is identified as one of the most bio-compatible materials [18], making it suitable as SHG probes in nonlinear microscopy for cell and tissue imaging. BTO NPs were also exploited as “Second Harmonic Radiation IMaging Probes (SHRIMPs) and used in live-cell imaging [19], drug delivery and conjugation of biomolecules [20]. With the help of digital phase conjugation, the BTO SHRIMP was used as a beacon to image through a turbid layer [21,22].

On the imaging front, defocused imaging has emerged as a versatile technique in many applications in recent years. Evaluating the orientation of a single molecule is one of the main applications of

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this technique [23–26]. With the help of the out-of-plane diffraction pattern, researchers have succeeded in tracking quantum dots and fluorescent nanoparticles in 3D inside living cells or any liquid environment [27,28]. For a nonlinear nanoparticle, a defocused image of the second harmonic (SH) light was shown to retrieve the 3D orientation of the nanocrystal, and reveal any deviation from the crystalline nature [29]. Earlier, researchers have observed coherent effects between the defocused SH patterns that emerged from two nanoparticles [30]. In this article, Sn-modified BaTiO₃ ferroelectric material, which is known for its enhanced ferroelectric and electro-caloric effect [31], has been synthesized. We investigate SHG from sub-micron size BTO particles. The out-of-plane diffraction patterns of SH light were analyzed thoroughly. We even show coherent effects in the SH light among multiple sub-micron size harmonic particle adjacents. Finally, another controlled experiment was performed with incoherent light to support our main result.

2. Sample preparation

We employed a conventional solid-state route for the preparation of poly-crystalline BaTi_{0.98}Sn_{0.02}O₃ compound. A small amount of Sn was doped with the aim of enhancing the nonlinear properties of BaTiO₃. Earlier, many groups observed doping-induced modifications in the optical nonlinearity of BaTiO₃ [32]. In the preparation process, BaCO₃, TiO₂, and SnO₂ are used as initial precursors for the synthesis of the said compound. Final sintering was done at 1300 °C for two hrs. X-ray diffraction (Cu K α) spectra revealed that the specimens were in a single phase within the detection limit (<2), and they exhibited a tetragonal structure (P4 mm space group) at room temperature. A small number of particles were dispersed in a methanol solution and ultra-sonicated for an hour towards declusterization. The decluttered nanoparticles were spin-coated onto an Indium Tin Oxide (ITO) coated glass plate, which aided in collecting the emission signal in transmission mode. Furthermore, its conductive nature supported the imaging in scanning electron microscopy (SEM) measurements. Au-Cr markers were lithographically inscribed onto the ITO to localize and identify individual particles. The SEM images showed a wide variation in the particles' sizes, extending from hundreds of nm to a few micrometers. Most of the particles ranged between 500 nm to 700 nm, which we refer to as sub-micron size harmonic particles.

3. Experimental setup

A custom-built wide-field nonlinear microscope was employed to collect second harmonic emission from the harmonic particles. In Fig. 1, we depict the schematic of the experimental design. An excitation beam from a Ti: Sapphire laser (specified pulse width 75 fs, repetition rate 80 MHz, wavelength 810 nm) was polarized in any desired plane using an achromatic half-wave plate. The beam was focused through an objective lens (NA 0.26) of 10X magnification, which offered a large excitation area for the simultaneous collection of second harmonic emission from a large area. Moreover, we underfilled the back aperture of the objective which offered a large excitation area at the focal plane for the simultaneous collection of second-harmonic emissions from a large area. The low magnification allows for an approximation of the excitation beam to be a plane wave [22]. The ITO-coated glass plate was positioned on an XYZ translation stage. In transmission, we collected the emitted second harmonic (SH) light and the fundamental light by another objective (NA 0.55) of 100X magnification, mounted on a translation stage. Next, the SH signal was filtered out by using two colored glass filters (transmission band 360 nm–580 nm) and a short-pass filter (transmission band 392 nm–725 nm) and finally imaged on an electron-multiplying charge-coupled device (EMCCD, Andor iXon Ultra 897), electrically cooled to –85 °C to restrain the thermal noise. The inset of Fig. 1 depicts two images. The left image exhibits the SEM image of an isolated single BTO particle of size around 630 nm, and the corresponding SH emission from the particle is depicted on the right side image.

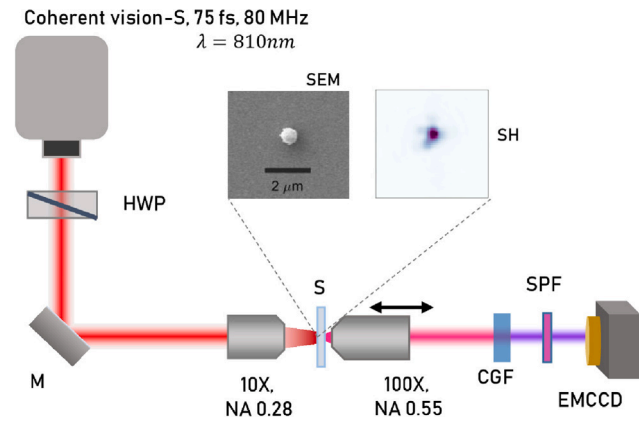


Fig. 1. Schematic of the experimental setup, Notations, HWP: Half wave plate, M: Mirror, S: Sample, CGF: Colored glass filter (Thorlabs, FGB39S, transmission band 360–580 nm), SPF: Short pass filter (Semrock, FF01-745/SP-25, transmission band 392–725 nm), EMCCD: Electron-multiplying charge-coupled device (Andor, iXon Ultra 897). Inset (left), Scanning electron micrograph (SEM) image. Inset (right), corresponding SH emission.

4. Results

Two control experiments were carried out to ascertain the SH nature of the emission, and to rule out fluorescence or photoluminescence. First, we investigated the output spectrum using a fiber-coupled spectrometer, which is plotted in Fig. 2(a). A distinct peak at approximately 405 nm (second harmonic for the fundamental wavelength of 810 nm) was observed, corresponding to the input laser wavelength of 810 nm. No other prominent peaks were noted elsewhere, nor any excessive broadening of the 405 nm peak was seen. It is important to note that, for this graph, we removed one of the color glass filters during the experiment. This adjustment permitted some of the fundamental photons to enter the spectrometer, facilitating the plotting of the fundamental and second harmonic on the same graph. Next, we measured the SH intensity as a function of the power of the input laser, as shown in Fig. 2(b). The blue discs are the experimental data, and the orange line is the power-law fit. We observed that the exponent turned out to be 2.03, which is in excellent agreement with the ideal value of 2 which symbolizes the well-known quadratic power dependence of SH intensity. Fig. 2(c) shows the data in a log–log scale showing a linear behavior with a slope of 2.03.

In a wide-field imaging system, a point-like object at the image plane looks like a spot with diameter $\sim \frac{1.22\lambda}{NA}$, where NA is the numerical aperture of the imaging objective lens. As we move the focal plane of the objective from the object, the diffraction pattern of concentric rings arises, and the radius of the diffraction rings increases with the distance between the object and the objective's focal plane (d) [33]. While this result has been shown for fluorescent particles, our experimental sample and setup offer the possibility to investigate the same utility for second-harmonic particles. With the help of markers, we compared the SEM and optical images of the second-harmonic particles. Fig. 3(a) shows such an identified single particle well-isolated from its neighborhood. In Fig. 3(b), we show the diffraction pattern of SH emission (defocused out-of-plane image) emerging from a single particle ($d = 30 \mu\text{m}$). We notice clear and radially symmetric concentric rings. On the other hand, Fig. 3(c) shows a close cluster of particles isolated from others. The diffraction from this cluster, shown in Fig. 3(d), shows the same rings, but with a clear distortion that evidently arises out of the clustered nature. The image does not exhibit signatures of interference. Importantly, these sources allowed us to investigate the utility of the SH nanoparticles for tracking in three-dimensions.

When the nanoparticle diffuses into the depths of a sample, the defocused imaging method traces the diameter of the diffraction ring.

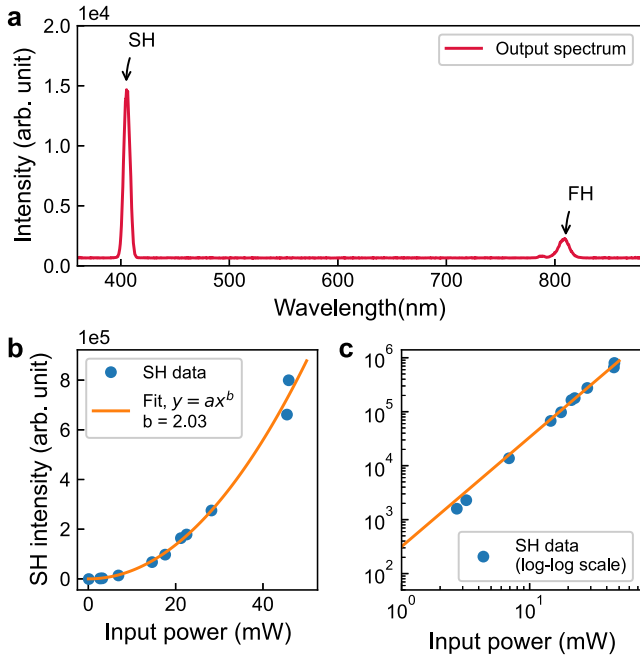


Fig. 2. (a) Emission spectrum shows a clear peak at 405 nm (second harmonic wavelength for the fundamental wavelength of 810 nm). (b) Output SH intensity plotted with input power variation (blue discs). Fit to the data (solid orange line) indicates a quadratic nature. (c) SH intensity plot in log-log scale showing linear behavior with a slope of 2.03, very close to the expected value of 2.

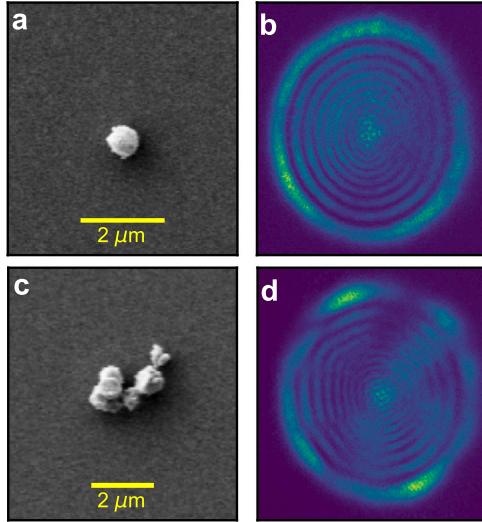


Fig. 3. Scanning electron micrograph images of two different types of particle assembly, (a) single BTO particle, (c) ensemble of few BTO particles, (b) and (d) are the out-of-focus diffraction patterns of the SH light emerging from particle (a) and (c) respectively.

We measured the radius of the outermost ring while varying d for different particles and plotted them in Fig. 4. Clearly, the fit (solid orange line) on the data points (blue discs) reveals the linear dependence of the diameter on the distance between the object and the objective's focal plane. This observation is consistent with the earlier findings on fluorescent nanoparticles [27], certifying that it is possible to track single HNP in environments such as living cells.

Next, we address the evolving formation of a speckle pattern as the number of constituent sources increases. Earlier, interference fringes were reported by Baumner et al. where SH emission from a pair of Potassium-Titanyl-Phosphate (KTP) nanoparticles was excited by

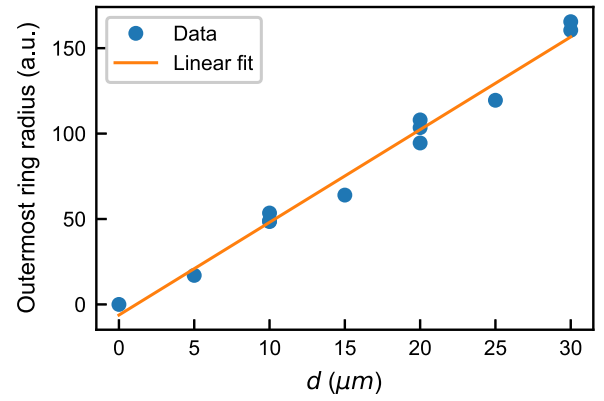


Fig. 4. The radius of the outermost ring (blue discs) is plotted against the distance between the object and the focal plane of the objective lens (d). The experimental data follows a linear behavior as observed from the fit (orange line).

evanescent fields [30]. Similarly, interference fringes were observed in the SH scattering patterns generated from single twinned and twin-free rods of ZnO [34]. Here, we present our observations with multi-particles and trace the evolution of a speckle pattern generated out of a random configuration. We note that speckle patterns are of immense fundamental interest in mesoscopic optics, and in particular, speckle from nonlinear random media has begun to be explored thoroughly in recent years [35–38]. By using various SEM images, we identify regions with two, three, or more particles that are well-separated. Fig. 5(a) depicts the SH image of two particles (separation ~ 5 μm) while the focal plane of the objective lens was at the substrate plane ($d = 0$). Fig. 5(b), (c) and (d) are the defocused images taken at $d = 10, 20, 30$ μm respectively. We noticed that as soon as the two patterns overlap, interference fringes emerge in the overlapped region. In Fig. 5(e) we present SH emission from four adjacent emission centers and the defocused images taken at $d = 5, 15, 25$ μm are shown in Fig. 5(f–h) respectively. Clearly, the overlap in the diffraction patterns immediately shows fringing. Fig. 5(i), (j), (k), and (l) show exemplary images formed from interference from two (i and j), three (k), and a large-area random cluster (l) of emission centers. The image in Fig. 5(l) essentially represents the classic speckle pattern from a random ensemble of second-harmonic emitters. The reason behind interference fringes is quite straightforward. Here, we excite those particles simultaneously with the same fundamental laser, and at the same time, unlike fluorescence, SH emission is a coherent process. As a result, the coherent superposition of defocused radiation patterns of SH light creates interference fringes.

To rule out any alignment error, we performed another experiment. First, incoherent light (essentially white light from a halogen source) was focused on the particles by the 100X objective. Then, we collect the back-scattered light from the particles with the same objective and finally image that on the EMCCD. Similar diffraction patterns emerged as we translated the objective lens's focal plane from the substrate plane. In Fig. 6, we show diffraction patterns emerging from multi-particles. As expected, we could not observe any interference fringe at the overlapped regions as the scattered light was incoherent.

5. Conclusion

In conclusion, we have synthesized Sn-doped BaTiO₃ (BTO) particles whose size varies from hundreds of nm to a few micrometers. We analyzed second-harmonic emissions from a single and an ensemble of BTO particles. With defocus, a diffraction pattern of SH light appeared, which helped identify the particle assembly's nature. When two or more neighboring particles were excited simultaneously and translated the objective, we observed interference fringes at the overlap regions

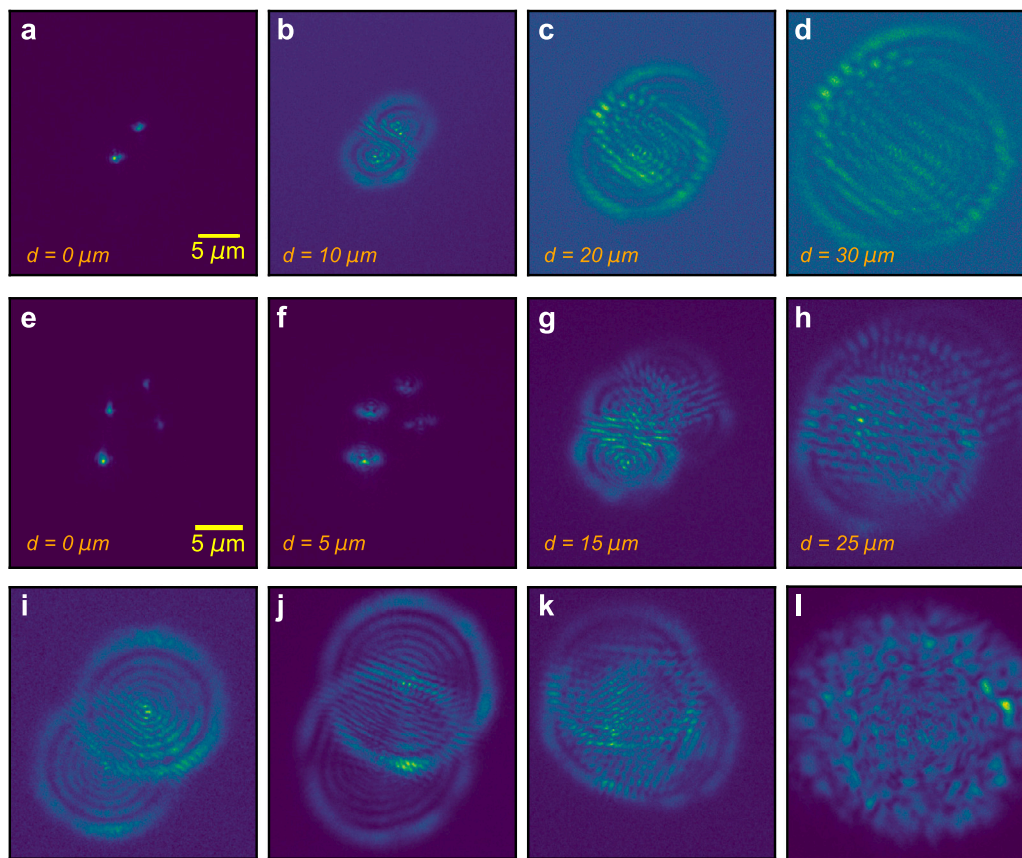


Fig. 5. (a–h) Focused to defocused images of second harmonic light for different collection objective positions. Here d refers to the distance between the object and the focal plane of the objective lens. [(a–d): two particles, (e–h): four particles]. (i, j) different sets of two particles, (k) three particles, and (l) multi-clustered particles.

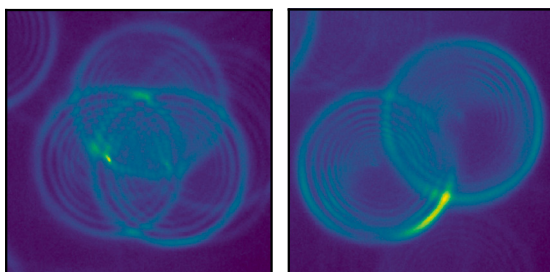


Fig. 6. Out-of-focus diffraction patterns of the backscattered incoherent light. It is clearly visible that no fringe is developed at the overlapped regions.

of their diffraction pattern. These fringes are essentially due to the coherent nature of SH emission. On the contrary, fringes were absent in case of incoherent light illumination. We noticed that the interference fringes' probability decreased where diffraction patterns appeared from a cluster of particles. Out-of-plane imaging of second-harmonic light can help in 3D tracking a harmonic nanoparticle in a living cell or liquid environment where both linear and nonlinear particles are present. In the last decade, harmonic holography has been employed to reconstruct inhomogeneous nonlinear structures buried in scattering media [39–43], wherein a reference second harmonic (SH) beam is utilized in the reconstruction process. It would be particularly interesting if the properties of inter-particle interference could be leveraged to access the unknown nonlinear structures inside a scattering medium, eliminating the need for a reference SH beam. Nevertheless, we believe that our study may contribute to advancements in deep tissue imaging.

CRediT authorship contribution statement

Rabisankar Samanta: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Sanjay K. Upadhyay:** Writing – review & editing, Resources. **Sushil Mujumdar:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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